

$$Y^* = \alpha + X\eta + \gamma + \sum_{\tau} D_{\tau}(\Delta\alpha_{\tau} + X\Delta\eta_{\tau} + \Delta\gamma_{\tau}). \quad (1)$$

0.0.1 Why pooling across school types biases logit estimates

In several applications it is tempting to pool all schools and impose common slope parameters (including the pandemic-year interaction slopes) across heterogeneous *school types*. This restriction is innocuous in linear models under standard exogeneity assumptions, but in nonlinear likelihood models it generally produces *misspecification* and hence inconsistent slope estimates when the conditional response function differs systematically by type.

Let $s \in \mathcal{S}$ denote school type (e.g. comprehensive/grammar/independent) and let $s(j)$ denote the type of school j . Consider the type-heterogeneous logit analogue of Equation (1) in which the index coefficients are allowed to differ by type:

$$\Pr(Y_{i,j,k,t} = 1 \mid X_i, j, k, t) = F\left(z_{i,j,k,t}(\theta_{s(j)})\right), \quad (2)$$

where θ_s collects all parameters relevant for that type (including the baseline mapping from X_i to grades and the pandemic-year interaction terms). To make the argument transparent, write a generic regressor vector as $x_{i,j,k,t}$ and a generic slope subvector as β_s so that, suppressing fixed effects for notational clarity, $\Pr(Y = 1 \mid x, s) = F(x'\beta_s)$.

Proposition 1 (Pseudo-true pooled logit and slope bias). Suppose the data are generated by $\Pr(Y = 1 \mid x, s) = F(x'\beta_s)$ with at least two school types s such that $\beta_s \neq \beta_{s'}$ on a set of positive probability, and let $\hat{\beta}^P$ denote the MLE from the *pooled* logit that (incorrectly) imposes a common slope β (i.e. estimates $F(x'\beta)$). Under standard regularity conditions for misspecified MLE, $\hat{\beta}^P \xrightarrow{P} \beta^P$, where the probability limit β^P solves the pooled population score equation

$$0 = \mathbb{E}[x(\mathbb{E}[Y \mid x, s] - F(x'\beta^P))] = \mathbb{E}[x(F(x'\beta_s) - F(x'\beta^P))]. \quad (3)$$

Define the type-weighted mean slope $\bar{\beta} := \mathbb{E}[\beta_s]$ and let $f(u) := F(u)(1 - F(u))$ denote the logit derivative. A first-order expansion of Equation (3) around $\bar{\beta}$ yields

$$\beta^P = \bar{\beta} + \beta_{\text{bias}}, \quad \text{with} \quad \beta_{\text{bias}} \approx \left(\mathbb{E}[f(x'\bar{\beta})xx']\right)^{-1} \mathbb{E}[f(x'\bar{\beta})xx'(\beta_s - \bar{\beta})]. \quad (4)$$

Unless β_s is constant across types (or heterogeneity is orthogonal to x in the specific sense that the right-hand expectation vanishes), $\beta_{\text{bias}} \neq 0$ and the pooled slope does not recover any type-specific causal response.

Derivation of Equation (4). Using $\mathbb{E}[Y \mid x, s] = F(x' \beta_s)$, the pooled score takes the form in Equation (3). Apply a Taylor expansion around $\bar{\beta}$: $F(x' \beta_s) \approx F(x' \bar{\beta}) + f(x' \bar{\beta}) x' (\beta_s - \bar{\beta})$ and $F(x' \beta^P) \approx F(x' \bar{\beta}) + f(x' \bar{\beta}) x' (\beta^P - \bar{\beta})$. Plugging into Equation (3) and collecting terms implies $\mathbb{E}[f(x' \bar{\beta}) x x'] (\beta^P - \bar{\beta}) \approx \mathbb{E}[f(x' \bar{\beta}) x x' (\beta_s - \bar{\beta})]$, which rearranges to Equation (4).

Corollary 1 (Correction required for unit-level treatment effects).

Let θ^P denote the pseudo-true pooled parameter vector (the probability limit of the pooled MLE under misspecification). Let $\tau_{i,j,k,\tau}(\theta)$ denote the unit-level grading-method effect implied by Equation (1) for pandemic year τ , i.e. the change in success probability induced by switching D_τ from 0 to 1 holding (X_i, j, k) fixed:

$$\tau_{i,j,k,\tau}(\theta) := F(z_{i,j,k,t} \mid D_\tau=1) - F(z_{i,j,k,t} \mid D_\tau=0). \quad (5)$$

If the true data-generating process features type-heterogeneous parameters $\theta_{s(j)}$ as in Equation (2), then in general

$$\tau_{i,j,k,\tau}(\theta^P) \neq \tau_{i,j,k,\tau}(\theta_{s(j)}), \quad (6)$$

because the unit effect is a nonlinear functional of the full index parameters. Therefore, some *correction* is required to consistently estimate unit-level treatment effects: a sufficient correction is to either (i) estimate Equation (1) separately within each school type s (so that $\hat{\theta}_s \xrightarrow{P} \theta_s$), or (ii) equivalently, estimate a single pooled likelihood that fully saturates type heterogeneity by interacting school-type indicators with all slope terms (including the pandemic-year interactions). Under either correction, $\hat{\tau}_{i,j,k,\tau} = \tau_{i,j,k,\tau}(\hat{\theta}_{s(j)})$ is consistent for $\tau_{i,j,k,\tau}(\theta_{s(j)})$.